

Generation Interconnection System Impact Study Report

For Queue Project Q19-1221

Sinai 138 kV
April 27, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1221. The Interconnection Customer proposes a 262.0 MW solar photovoltaic facility interconnecting at the Sinai 138 kV substation in the Virginia area. The request was received on June 08, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 330 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$31,209,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
SINAI - HALIFAX 138.0 kV	VENWIND - SINAI 138.0 kV	131.0	338	55.3
SINAI - VENWID 138.0 kV L	SINAI - HALIFAX 138.0 kV	129.9	344	54.6
SYLVAN LKE - GRS LYON 138.0 kV Line	CAPE HORN - HUBEN 138.0 kV	129.7	245	28.6
HALIFAX - VENWIND 138.0 kV	VENWIND - SINAI 138.0 kV	121.3	331	40.3
HUBEN - SYLVAN LAKE SUBSTATION 138.0 kV	HUBEN - CAPE HORN 138.0 kV	121.3	268	28.6
VELVA WIND INTERCONNECTION - GRS LYON 138.0 kV	HUBEN - CAPE HORN 138.0 kV L	119.6	262	28.6
HUBEN - VENWIND 138.0 kV	BEAR - CLOVER 345.0 kV Line	115.2	462	45.6
WELCO - PINE RIVER ALTE 138.0 kV L	VENWIND - HUBEN 138.0 kV	107.9	367	72.9
SHENANDOAH - WALWORTH 138.0 kV	BEAR - CLOVER 345.0 kV	106.1	345	5.4
AIBONITO PDS - SHENANDOAH 138.0 kV L	BEAR - CLOVER SUBSTATION 345.0 kV L	105.4	367	5.4
JETERSVILLE - BLUFF VIEW 138.0 kV line	BEAR - CLOVER 345.0 kV	105.0	236	6.6
AIBONITO PDS SUBSTATION - INTEL 138.0 kV	AIBONITO PDS - S CRK 138.0 kV L	103.9	277	7.8
BLUFF VIEW - SELMA 138.0 kV	CLOVER - BEAR SUBSTATION 345.0 kV	102.4	231	6.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HALIFAX - SINAI 138.0 kV	Rebuild line with higher-capacity conductor	\$6,136,000	40% (\$2,428,000)
SINAI - VENWIND 138.0 kV	Rebuild line with higher-capacity conductor	\$2,942,000	40% (\$1,163,000)
SYLVAN LKE - GRS LYON 138.0 kV Line	Rebuild line with higher-capacity conductor	\$20,647,000	25% (\$5,155,000)
HALIFAX - VENWIND 138.0 kV	Rebuild line with higher-capacity conductor	\$2,734,000	36% (\$982,000)
SYLVAN LAKE - HUBEN SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$7,700,000	25% (\$1,922,000)
VELVA WIND INTERCONNECTION - GRS LYON 138.0 kV	Rebuild line with higher-capacity conductor	\$5,927,000	25% (\$1,480,000)
VENWIND - HUBEN 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,732,000	36% (\$616,000)
WELCO - PINE RIVER ALTE 138.0 kV L	Reconductor line	\$3,599,000	39% (\$1,416,000)
WALWORTH - SHENANDOAH 138.0 kV	Reconductor line	\$7,649,000	73% (\$5,587,000)
AIBONITO PDS - SHENANDOAH 138.0 kV L	Reconductor line	\$7,121,000	73% (\$5,202,000)
JETERSVILLE - BLUFF VIEW 138.0 kV line	Reconductor line	\$3,644,000	46% (\$1,676,000)
INTEL - AIBONITO PDS 138.0 kV	Reconductor line	\$7,551,000	42% (\$3,205,000)
BLUFF VIEW - SELMA 138.0 kV	Reconductor line	\$821,000	46% (\$377,000)
	Total allocated to project		\$31,209,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.